Data Visualization Project

Analysis On SuperStore In U.S.

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Project Overview

This project analyzes the "Superstore Dataset" (5,000 orders, 2014–2017) using Tableau to identify growth drivers and profitability leaks. The goal is to evaluate sales trends and provide actionable strategies for profit improvement.

Data Cleaning and Preprocessing

The initial CSV suffered from structural corruption, specifically unquoted extra commas and nested data in the "Product Name" field. This caused a column shifting issue in 21 rows, pushing critical values (Sales, Profit) into incorrect "Unnamed" columns. I manually realigned these 21 rows and standardized the dataset into a 21-column format to ensure data integrity for analysis.

Methodology

Tableau was chosen as the tool for this project's data visualization and analysis phase. Its interactive interface allowed us to quickly spot outliers and perform "drill-down" analysis across regions and categories, transforming raw data into a dynamic view of business performance.

Highlights

The key findings of this report are structured into three major highlights, each derived from our three Tableau dashboards: establishing the current status via the Overview Dashboard, the Sales & Profit Dashboard, and Category Dashboard

Overview Dashboard Analysis (Appendix A)

Based on the Overview Dashboard, Superstore demonstrated significant growth in business scale between 2014 and 2017, but also revealed several hidden operational risks:

<u>Key Performance Indicators</u>

- **KPIs:** Sales grew steadily from $483k (2014) to $728k (2017). However, a **declining AOV (Average Order Value)** suggests growth is driven by sheer volume rather than value—a potential long-term risk.

<u>Sub-category Analysis</u>

- **Copiers** are the efficiency leaders ($49k sales / $17k profit), while **Tables** are the primary loss-maker (deep red). **Phones and Chairs** exhibit a "High Sales, Low Profit" trap, likely due to excessive discounting.

- Therefore, these products show a **"high sales, low profit" pattern.** We suspect this is due to **heavy discounting**, which increases sales but destroys the profit margin.

Market Distribution

- **Top 3 States (Over-dependence):** Sales are mostly concentrated in: California: $131,330,New York: $71,379,Texas: $41,643

- **Low Market Penetration:** Only 6 states have sales between $20,000 and $30,000. Most states have sales under $10,000.

- **Risk**: Extreme dependence on California, New York, and Texas creates vulnerability. **Expanding market penetration** in mid-tier states is recommended.

Sales & Profit Dashboard Analysis (Appendix B)

Following the discovery of low-profit margins in the overview, the Sales & Profit Dashboard provides a deep-dive diagnosis into the root cause: Over-discounting.

Discount Bracket Analysis

- **The 20% Profitability Red Line:** Analysis proves profitability remains stable only when discounts are **below 20%**. Beyond this threshold, profit margins collapse, particularly in the "Over 50%" bracket.

- Therefore, this proves that our current aggressive promotion strategy is the primary driver of profit decline.

Scatter Plot Analysis

- **Negative Correlation:** The scatter plot reveals a strong negative correlation between discount rates and profit margins; as discounts increase, profitability consistently declines.

- **Concentration of Losses:** High-discount orders are heavily clustered in the "Loss Zone" (below the 0% profit line), proving that excessive discounting is the primary driver of systematic losses.

Regional Risk & City Drill-down

- **Performance Contrast:** Using **"TOP/BOTTOM 5"** filters, we clearly identified the stark gap between the best and worst-performing states.

- **Profit Leaders: California** and **New York** remain our strongest markets, maintaining high profitability with average discounts of only **10-15%**.

- **Loss-making "Red Zones":** In contrast, states such as **Texas**, **Ohio**, and **Pennsylvania** suffer the heaviest losses, The most loss-making cities have average discounts exceeding **40%.**

- **Strategic Risk:** We are currently **"buying sales"** at a significant loss in these Bottom 5 states. We recommend an immediate **20% discount cap** for these "Red Zone" cities and a re-evaluation of pricing strategies for low-margin sub-categories.

Category Dashboard Analysis (Appendix C)

This dashboard provides a granular analysis of sales growth trends, profitability by category, and individual product performance, identifying the true engines of Superstore's revenue.

Year-over-Year Sales Growth Trends

The dataset reveals significant fluctuations in sales expansion across the analyzed period:

- **2015:** Sales grew by an average of **12.3%** compared to the previous year, showing steady initial progress.
- **2016:** The business experienced explosive growth, with an average increase of **64.5%**, marking a major market breakthrough
- **2017:** Growth stabilized at **19.7%**, suggesting the market is reaching maturity and the strategy should shift from "volume expansion" to "margin optimization."

High Volume vs. High Profitability (The 12.72% Benchmark)

By comparing sub-categories against the **overall average profit margin of 12.72%**, we identified a clear disconnect between sales volume and profitability:

- **The High-Volume Trap: Phones** and **Chairs** generated the highest total sales at **$330,007** and **$328,449** respectively. However, Phones yielded an average profit margin of only **11.92%**, falling below the company average. This confirms that high-revenue items are often weighed down by aggressive discounting or high operational costs.

- **High-Value Performer:** In contrast, **Accessories** and **Copiers** are the most efficient categories. While their sales were lower ($167,380 and $149,528 respectively), their average profit margins reached an impressive **21.82%** and **31.72%**, far exceeding the corporate benchmark.

Identification of Winning Products

The analysis highlights specific products that act as primary profit drivers:

- **Copiers - The "Winning Product":** The **Canon imageCLASS 2200 Advanced Copier** stands out with exceptionally high sales and margins, making it the single most valuable item in the inventory.

- **Accessories - Top 3 Performers:** The success of this high-margin category is driven by:

    1. *Plantronics Cs510 - Over-the-head monaural wireless headset system*

2. *Logitech p710e mobile speakerphone*
3. *Plantronics savi w720 multi-device wireless headset system*

Strategic Recommendations

Based on the insights from all three dashboards, we propose the following strategies to improve Superstore's profitability in 2026:

- **Shift Marketing Focus:** Reallocate advertising budget from low-margin categories like Chairs and Tables toward high-margin "Stars" such as **Copiers** and **Accessories**.

- **Optimize the "Phone" Category:** Since Phones have the highest sales volume ($330,007) but sub-average margins (11.92%), the store should implement a **"Bundle Strategy."** Instead of direct discounts on phones, offer them at full price bundled with high-margin accessories (e.g., Plantronics headsets) to pull the overall transaction margin above the 12.72% line.

- **Enforce a 20% Discount Cap:** Data from the Sales & Profit Dashboard proves that discounts exceeding 20% destroy value. A strict price floor must be established, particularly in "Red Zone" states like Texas and Ohio.

- **Leverage Winning Products:** Use the success of the **Canon imageCLASS 2200** as a case study to identify high-value corporate clients, offering them loyalty programs that encourage repeat purchases of high-margin consumables (ink, paper, and accessories).

Effectiveness of the Visualizations

In this project, Tableau visualizations served as critical diagnostic tools rather than mere displays, allowing us to move from raw data to actionable hypotheses.

- **Hypothesis Generation via Geographic Maps:** By using a **Filled Map**, we instantly contrasted Sales volume with Profit color density. This led to the hypothesis that *"market scale does not equal market health,"* specifically observing that high-sales states like Texas were in a "Red Zone" of deep losses.

- **Root Cause Analysis via Scatter Plots:** The **Scatter Plot** was instrumental in proving the correlation between discount rates and profit margins. By adding a trend line, we mathematically visualized the "20% Profitability Red Line," confirming that aggressive discounting was the primary driver of profit collapse.

- **Strategic Precision via Drill-down Analysis:** We employed a systematic **Drill-down** approach. First, a Scatter Plot identified **Copiers** as a high-potential sub-category. We then transitioned to a **Sorted Bar Chart** to isolate the specific "Winning Product." This visual synergy allowed us to precisely identify the **Canon imageCLASS 2200 Advanced Copier** as the top profit contributor.

- **Benchmarking via Reference Lines:** Implementing a **Reference Line for the Average Profit Margin (12.72%)** turned complex tables into a clear "above or below" benchmark, instantly categorizing products like **Phones** as high-volume underperformers.

Challenges and Limitations

- **Technical Hurdles:** Significant CSV corruption, including extra commas and nested data, caused 21 rows to shift. This required manual realignment to ensure data integrity for visualization.

- **Data Constraints:** As the dataset ends in 2017, it lacks modern e-commerce trends and post-pandemic consumer behaviors relevant to 2026. Additionally, missing "Shipping Cost" data limits the profitability analysis for bulky furniture.

- **Future Improvements:** Integrating real-time inventory or social media sentiment analysis would provide a more holistic view of product quality and customer satisfaction.

Future Work

- **Description of Additional Ideas:**

    1. **Predictive Analytics:** Implementing a Linear Regression model or Time Series forecasting to predict 2026 sales based on seasonal cycles.

    2. **Customer Segmentation (RFM Analysis):** Categorizing customers based on Recency, Frequency, and Monetary value to identify high-value corporate clients for the "Winning Product" loyalty program.

- **Reasons for Non-Implementation:** These approaches were not implemented due to **time constraints** and the project's specific focus on **Descriptive Visualization** (understanding the past) rather than Predictive Modeling (forecasting the future).

Conclusion

This project transformed 5,000 records into a strategic roadmap. By cleaning the corrupted data and building three dashboards, we successfully identified profit leakages and revenue drivers.

1. **Discount Control:** Discounts exceeding 20% are the single largest threat to profit, especially in the Furniture category.

2. **Profit Efficiency:** High volume does not mean high value. Sub-categories like **Accessories** and **Copiers** are far more efficient than high-sale items like **Phones**.

3. **Product Strategy:** The identification of the **Canon imageCLASS 2200** as a winning product provides a clear template for future high-margin inventory expansion.

Reference

https://www.kaggle.com/datasets/vivek468/superstore-dataset-final/data

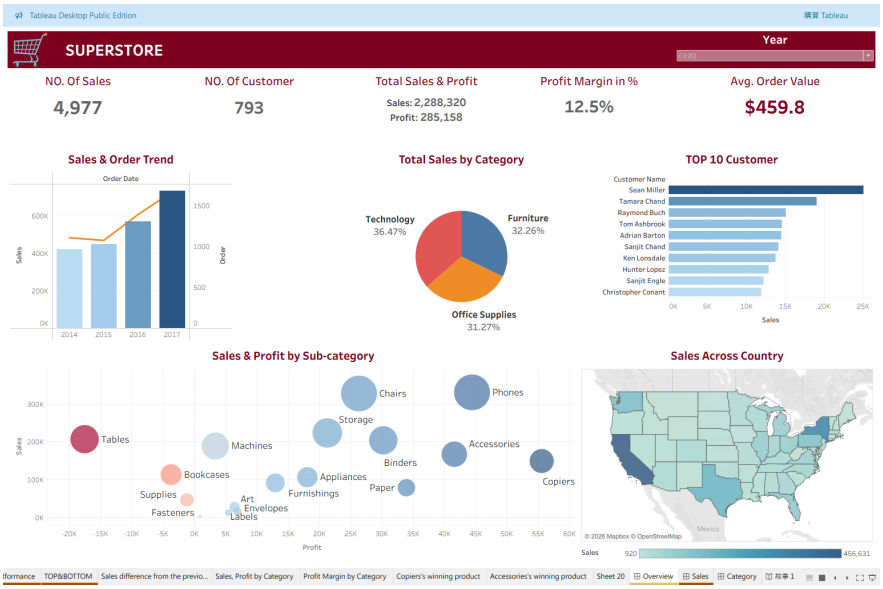
Distribution of Work

Dashboard : Ian and Larry

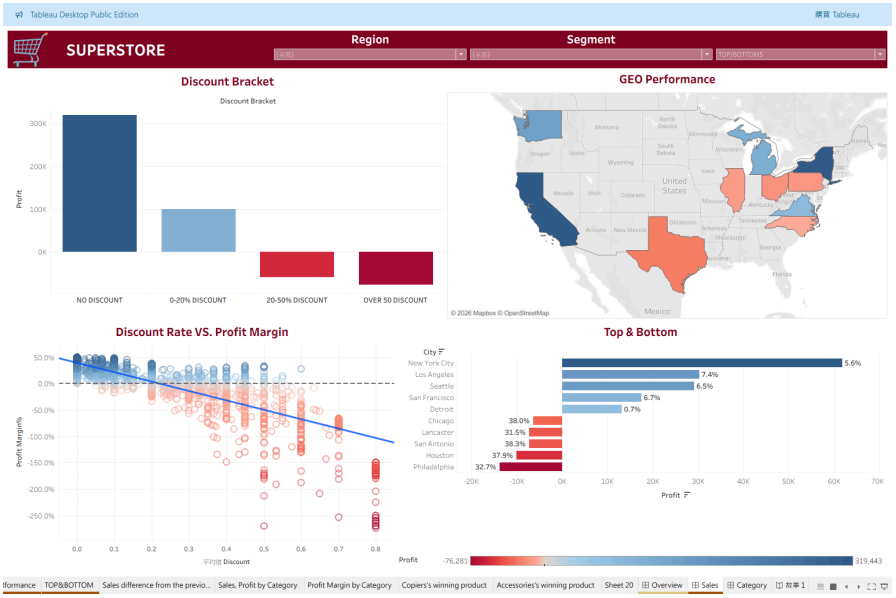
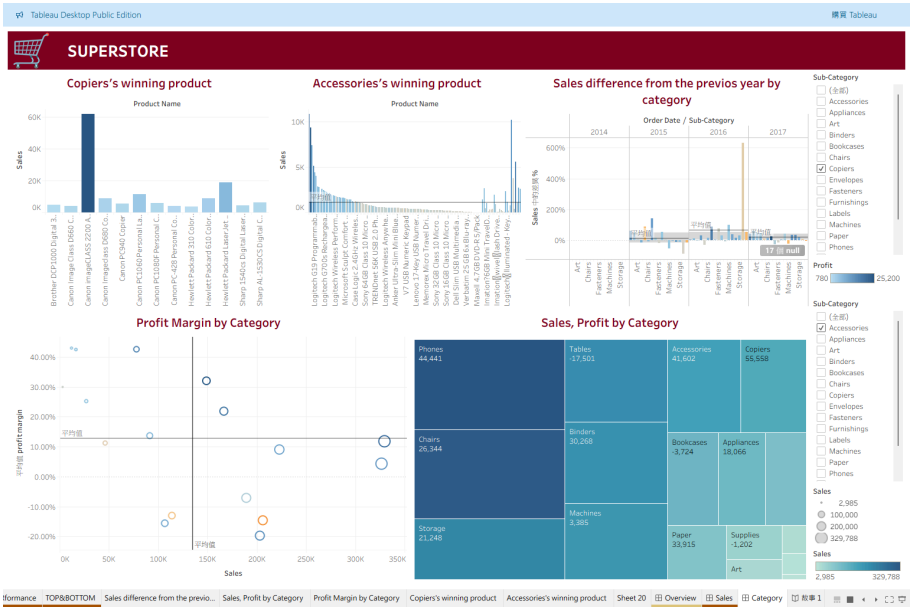
Report: Ian and Larry

Appendix

A. OverView Dashboard



B. Sales and Profit Dashboard



C. Category Dashboard